

SODU BACKEND

RFQ MODULE

Integration Technical Report

Complete API reference, data structure overview, and deployment guide for the Replit RFQ module integration with the Sodu production backend.

DEPLOYED

LIVE

v1.0

SERVER

VPS · 72.62.234.67

BASE URL

testdeployfornetservex.online/sudobe

PROCESS

PM2 · sudobe (id 12)

DATE

June 2026

1.1 Backend Stack		
Layer		
Runtime	Node.js	Express 5.x — async/await throughout
Database / BaaS	Swell	All data in Swell — no local SQL DB
Authentication	JWT + API Key	Admin: JWT cookie RFQ: X-RFQ-API-Key
File Storage	Swell Files API	Base64 upload !' CDN URL returned
2FA	Twilio	WhatsApp OTP (SMS fallback)
Process Manager	PM2	Process id 12 · fork mode
Hosting	VPS	72.62.234.67 — nginx reverse proxy
Audit Logs	Swell CMS	/content/activity-logs
Shipping ETA	SeaRates API	Container ETA calculation
Container Tracking		

1.2 Key Swell Collections

Swell Endpoint

/accounts (factory_name = null)

Customers / Buyers

content.factory_name = null

/accounts (factory_name != null)

Factories / Suppliers

content.factory_name set

/orders

All orders

content.order_status, content.rfq_id

/products

Product catalog

content.factory_id links to account

/categories

Product categories

slug, parent_id, top_id

/content/cms-users

Admin panel users

content.role = super_admin | operator

/content/activity-logs

Audit trail

content.action, content.resource_id

/content/shipping-rates

Per-country rates

content.country_name (ISO 2)

/content/role-permissions

Dynamic RBAC

content.role_key, content.permissions

/:files

Uploaded files

1.3 User Roles & Permissions

Role

super_admin	Full access	All permissions — manage users, roles, all CRUD
operator	Limited access	View+edit orders, view customers/factories/products, view QA

2.1 Customer (Swell /accounts where factory_name = null)

Customer Object (/accounts)

```
{
  "id":      "swell_account_id",
  "email":   "buyer@company.com",
  "first_name": "Ahmed",
  "last_name": "Hassan",
  "phone":   "201234567890",
  "content": {
    "verified":      true,
    "vetted":        true,
    "company_name":  "Al Nour Trading",
    "country":       "Guinea",
    "registration_number": "GN-REG-001",
    "role":          "buyer"
  }
}
```

2.2 Factory / Supplier (Swell /accounts where factory_name != null)

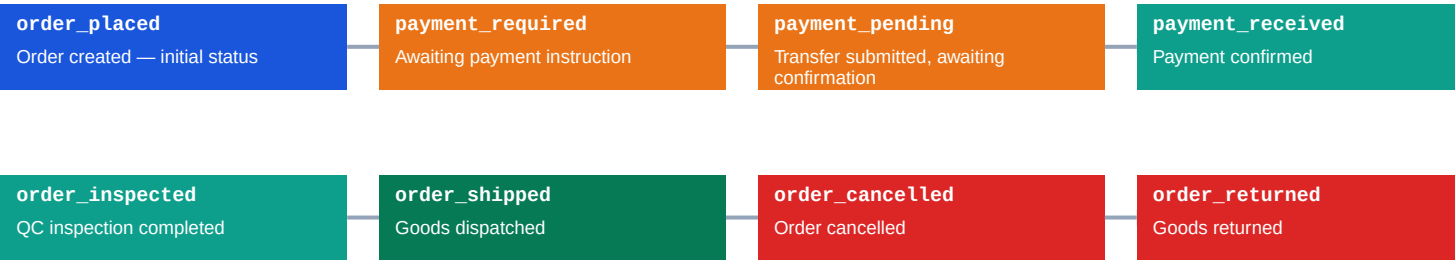
Factory Object (/accounts - sensitive fields stripped for RFQ)

```
{
  "id":      "swell_account_id",
  "email":   "factory@example.com",
  "name":    "Cairo Gypsum Industries",
  "content": {
    "factory_name":  "Cairo Gypsum Industries",
    "country":       "Egypt",
    "city":          "Cairo",
    "lead_time":     { "min_days": 10, "max_days": 20 },
    "minimum_quantity": "1000",
    "verified":      true,
    "vetted":        true,
    "store_front_logo": { "url": "https://cdn.swell.store/..." }
  }
  // personal_id & certifications are STRIPPED from RFQ responses
}
```



```
{
  "id": "swell_order_id",
  "number": 2091,
  "account_id": "swell_customer_id",
  "items": [{ "product_id": "...", "quantity": 4200 }],
  "status": "pending",
  "content": {
    // Standard Sodu order fields
    "order_status": "order_placed", // see statuses below
    "date_of_shipping": "2026-08-01",
    "delivery_date": "2026-09-01",
    "transfer_id": "TRF-000123",
    "shipping_company_name": "...",
    "qc_required": true,
    // RFQ-specific fields (set by from-rfq endpoint)
    "rfq_id": "RFQ-1048",
    "approved_quote_id": "QUOTE-456",
    "source": "rfq_module",
    "product_name": "Moisture Resistant Gypsum Board",
    "quantity": 4200,
    "unit_price": 4.56,
    "total_price": 19152,
    "currency": "USD",
    "incoterm": "EXW",
    "selected_factory_id": "factory_account_id",
    "destination_country": "Guinea",
    "destination_port": "Conakry",
    "payment_terms": "100% in advance",
    "lead_time_days": 15
  }
}
```

2.4 Order Status Flow (stored in content.order_status)



2.5 Authentication Methods

A — Admin Panel (JWT)

Admin users are stored in Swell at /content/cms-users. On login (/api/auth/login), a JWT is issued (24h expiry) and set as an HTTP-only cookie. Protected routes read the cookie or the Authorization: Bearer <token> header. The JWT payload contains: { id, email, verified }.

B — RFQ Module (API Key)

The Replit RFQ module authenticates with a static API key in the request header:

```
Header (required on every RFQ API call)

X-RFQ-API-Key: sodu_rfq_9f3a7b2e1d4c8f6a0e5b3d7c2a9f1e4b

// Wrong key !' 403 Forbidden
// Missing header !' 401 Unauthorized
```

The server-side key is stored in /var/www/html/sudobe/.env as RFQ_SECRET_KEY. Share only with the Replit project as a secure environment variable. Rotate if compromised.

Base URL: <https://testdeployfornetserverx.online/sudobe/api/rfq>
All endpoints require header: X-RFQ-API-Key: sodu_rfq_9f3a7b2e1d4c8f6a0e5b3d7c2a9f1e4b

3.1 Endpoint Reference

Method			
GET	/me	Validate API key / ping server	API Key
GET	/customers	List customers (?search= ?limit=)	API Key
GET	/customers/:id	Single customer by Swell account ID	API Key
GET	/factories	List factories (sensitive fields stripped)	API Key
GET	/factories/:id	Single factory by Swell account ID	API Key
GET	/categories	All product categories (id + slug)	API Key
POST	/upload	Upload file !' returns { url, id }	API Key

POST

/orders/from-rfq

Convert approved RFQ !' real Swell order

API Key

GET

/orders/:id

Fetch order (RFQ-safe fields only)

API Key

3.2 GET /api/rfq/me — Connectivity Test

GET /me

```
// Request
GET /sudobe/api/rfq/me
X-RFQ-API-Key: sodu_rfq_9f3a7b2e1d4c8f6a0e5b3d7c2a9f1e4b

// Response 200
{
  "success": true,
  "message": "RFQ API key is valid",
  "system": "Sodu Backend",
  "version": "1.0"
}
```

3.3 GET /api/rfq/customers — Customer List

GET /customers

```
// Request
GET /sudobe/api/rfq/customers?search=Ahmed&limit=100
X-RFQ-API-Key: sodu_rfq_...

// Response 200
{
  "success": true,
  "count": 3,
  "data": [
    {
      "id": "swell_account_id",
      "email": "buyer@company.com",
      "first_name": "Ahmed",
      "phone": "201234567890",
      "content": { "company_name": "...", "country": "Guinea", "vetted": true }
    }
  ]
}
```

3.4 POST /api/rfq/orders/from-rfq !• Most Important Endpoint

This endpoint converts an approved RFQ client offer into a real Swell order. Once called, the normal Sodu order flow takes over (QC, payment, shipping, tracking). All RFQ metadata is stored in order.content for full audit trail.

Request Payload

POST /orders/from-rfq – Request

```
POST /sudobe/api/rfq/orders/from-rfq
Content-Type: application/json
X-RFQ-API-Key: sodu_rfq_9f3a7b2e1d4c8f6a0e5b3d7c2a9f1e4b

{
  // Required fields
  "customerId": "swell_account_id", // Swell account ID of the customer
  "rfqId": "RFQ-1048",
  "approvedQuoteId": "QUOTE-456",
  "productName": "Moisture Resistant Gypsum Board",
  "quantity": 4200,
  "unitPrice": 4.56,

  // Recommended fields
  "categoryId": "construction_materials", // from /api/rfq/categories
  "unit": "boards",
  "destinationCountry": "Guinea",
  "destinationPort": "Conakry",
  "selectedFactoryId": "swell_factory_account_id",
  "currency": "USD",
  "incoterm": "EXW",
  "leadTimeDays": 15,
  "paymentTerms": "100% in advance",
  "qcRequired": true,
  "shippingRequired": true,
  "attachments": [],
  "notes": "Converted from approved RFQ client offer.",

  // Optional – pass Swell product ID if known (improves order line detail)
  "productId": "optional_swell_product_id"
}
```

Success Response (201 Created)

POST /orders/from-rfq – Response 201

```
{
  "success":      true,
  "orderId":      "6876abc123def000",    // Swell order ID – store in RFQ DB
  "orderNumber":  2091,
  "status":       "Created",
  "trackingUrl":  "https://testdeployfornetserverx.online/sudobe/orders",
  "rfqId":        "RFQ-1048",
  "approvedQuoteId": "QUOTE-456",
  "customerId":   "swell_account_id",
  "warnings":     []    // or ["No matching Swell product found..."] if productId missing
}
```

Error Responses

HTTP

400	Missing required fields	"Missing required fields: customerId, rfqId, ..."
401	No API key header	"Missing X-RFQ-API-Key header"
403	Wrong API key	"Invalid RFQ API key"
404	Customer ID not in Swell	"Customer with id ..." not found in Sodu system"
404	Factory ID not in Swell	"Factory with id ..." not found in Sodu system"

400

Account is a factory

"Account ..." is a factory, not a customer"

500

Swell error

"Failed to create order from RFQ"

3.5 GET /api/rfq/factories

GET /factories – Response

```
// Returns factories with sensitive fields stripped:
// personal_id, certifications, price_sidenote are REMOVED

{
  "success": true,
  "count": 42,
  "data": [
    {
      "id": "swell_account_id",
      "name": "Cairo Gypsum Industries",
      "email": "factory@example.com",
      "content": {
        "factory_name": "Cairo Gypsum Industries",
        "country": "Egypt",
        "city": "Cairo",
        "lead_time": { "min_days": 10, "max_days": 20 },
        "verified": true,
        "vetted": true
      }
    }
  ]
}
```

3.6 POST /api/rfq/upload — File Upload

POST /upload

```
// Multipart form-data
```


API key auth

Every /api/rfq/* route requires X-RFQ-API-Key — 401/403 if absent/wrong

Sensitive data

personal_id, certifications, price_sidenote, otp stripped server-side

No order modification

RFQ module can only create orders — no PUT/DELETE on existing orders

Audit trail

Every from-rfq call logged to /content/activity-logs with rfq_id + quote_id

Customer isolation

from-rfq validates account is a customer (not a factory)

Factory bid privacy

Factory details returned without internal KYC/compliance docs

No margin exposure

price_sidenote (internal cost notes) stripped from all factory responses

Order audit fields

content.rfq_id, content.approved_quote_id, content.source on every RFQ order

5.1 What Replit Should Store (External IDs)

Field		
customer_id	GET /api/rfq/customers !' id	
		Swell account ID
factory_id	GET /api/rfq/factories !' id	
		Swell account ID
category_id	GET /api/rfq/categories !' id or slug	
		Used in from-rfq payload
product_id	Swell catalog (optional)	
		Pass to improve order line detail
created_order_id	POST /api/rfq/orders/from-rfq !' orderId	
		Store after successful conversion

5.2 Replit RFQ Data Model (Own Tables)

The Replit module should maintain its own database for RFQ workflow data. Do NOT duplicate customers, factories, products, or orders from Swell. Store only references:

Table	
rfq_requests	
	rfq_id, customer_id*, status, product_name, qty, ...
factory_invitations	
	rfq_id, factory_id*, invite_token, status
factory_quotes	
	rfq_id, factory_id*, unit_price, currency, lead_time, ...

quote_comparisons	rfq_id, [quote_ids], selected_quote_id
client_offers	rfq_id, client_id*, offer_price, validity_date
client_approvals	rfq_id, offer_id, approved_at, created_order_id*
rfq_status_history	rfq_id, old_status, new_status, changed_by, changed_at

5.3 RFQ Flow !' Order Creation

- 1 Customer/Admin creates RFQ in Replit
- 2 Admin cleans & invites factories (Replit)
- 3 Factories submit quotes via secure links (Replit)
- 4 Admin compares quotes, creates client offer (Replit)
- 5 Client approves offer (Replit)
- 6 Replit calls POST /api/rfq/orders/from-rfq !' Swell order created
- 7 Sodu normal flow continues: QC !' Payment !' Shipping !' Tracking

6. Deployment Details

Item	
Server	VPS 72.62.234.67
App directory	/var/www/html/sudobe
New files deployed	/var/www/html/sudobe/middleware/rfqAuth.js
	/var/www/html/sudobe/routes/rfqRoutes.js

Updated files

/var/www/html/sudobe/app.js

Env variables added

RFQ_SECRET_KEY + APP_BASE_URL (appended to .env)

PM2 process

sudobe (id 12) — restarted with --update-env

Live base URL

https://testdeployfornetservex.online/sudobe/api/rfq

Status check

GET /api/rfq/me !' 200 OK ' (verified post-deploy)

RFQ API Key

sodu_rfq_9f3a7b2e1d4c8f6a0e5b3d7c2a9f1e4b

6.1 Pre-production Checklist

- Set RFQ_SECRET_KEY in both server .env and Replit env to a strong random secret (openssl rand -hex 32)
- Add APP_BASE_URL=https://app.gosodu.com (or production domain) to server .env
- Rotate the key above — the one in this document was used for testing only
- Replit: store customer_id, factory_id, category_id from the RFQ API before calling from-rfq
- Optional: pass productId in from-rfq to attach a real catalog item to the order line
- Verify nginx proxies /sudobe !' localhost:5000 (already configured on VPS)

These are the existing APIs used by the admin panel and mobile app. The RFQ module should NOT call these directly — use the /api/rfq/* wrapper endpoints instead.

7.1 Admin Panel Routes (JWT cookie required)		
Method		
GET	/orders/api	List orders (paginated, filterable)
GET	/orders/api/:id	Single order + account + items
POST	/orders/api	Create order (multipart, file uploads)
PUT	/orders/api/:id	Update order
DELETE	/orders/api/:id	Delete order
GET	/customers/api	List customers
GET	/customers/api/:id	Single customer
POST	/customers/api	Create customer
PUT	/customers/api/:id	

	Update customer
DELETE	
/customers/api/:id	Delete customer
GET	
/factories/api	List factories
POST	
/factories/api	Create factory
PUT	
/factories/api/:id	Update factory
DELETE	
/factories/api/:id	Delete factory
GET	
/shipping-rates/api	List shipping rates
GET	
/logs/api	Activity audit logs
GET	
/permissions/api	Role permissions CRUD
GET	
/api/auth/me	Current admin user info
POST	
/api/auth/login	Admin login
GET	
/api/auth/logout	Admin logout

GET

/api/catalogue/all-products

All products grouped by factory

GET

/api/catalogue/search-products

Search products

GET

/api/catalogue/categories

Product categories

GET

/api/catalogue/subcategories/:id

Subcategories by parent ID

GET

/api/catalogue/productDetails/:id

Single product details

GET

/api/catalogue/factories

Public factory list

POST

/api/auth/register-user

Register customer (OTP flow)

POST

/api/checkout/update-order-status/:id

Update order status

GET

/api/checkout/get-order-details

Get order details with tracking

GET

/api/checkout/track

Track container by number

GET

/api/shipping-rates

Public shipping rates

POST

/api/account/addresses

Customer address management

8.1 Step-by-Step: Create Your First RFQ Order

Step 1 — Verify connectivity

Step 1 – Ping

```
curl https://testdeployfornetserverx.online/sudobe/api/rfq/me \
  -H "X-RFQ-API-Key: sodu_rfq_9f3a7b2e1d4c8f6a0e5b3d7c2a9f1e4b"

// Expected: { "success": true, "message": "RFQ API key is valid" }
```

Step 2 — Fetch data you need before creating an order

Step 2 – Fetch reference data

```
// Get customers
GET /api/rfq/customers

// Get factories
GET /api/rfq/factories

// Get categories
GET /api/rfq/categories

// Store the "id" fields – you will need them in Step 3
```

Step 3 — Convert approved RFQ to order

Step 3 – from-rfq call

```
curl -X POST https://testdeployfornetserverx.online/sudobe/api/rfq/orders/from-rfq \
  -H "X-RFQ-API-Key: sodu_rfq_9f3a7b2e1d4c8f6a0e5b3d7c2a9f1e4b" \
  -H "Content-Type: application/json" \
  -d '{
    "customerId": "<id from Step 2>",
    "rfqId": "RFQ-1048",
    "approvedQuoteId": "QUOTE-456",
    "productName": "Moisture Resistant Gypsum Board".
```

Step 4 – Store orderId

```
// Save "orderId" to your RFQ database (rfq_approvals table)
{
  "success": true,
  "orderId": "6876abc123def000",    !• store this!
  "status": "Created",
  "trackingUrl": "https://testdeployfornetservev.online/sudobe/orders"
}

// The Sodu admin panel will now show the order and manage it from here.
// No further action needed from the RFQ module for this order.
```

8.2 Environment Variables for Replit**Replit Environment Variables**

```
# Add these to your Replit .env / Secrets
SODU_RFQ_API_KEY=sodu_rfq_9f3a7b2e1d4c8f6a0e5b3d7c2a9f1e4b
SODU_BASE_URL=https://testdeployfornetservev.online/sudobe

# Usage in code
const headers = { "X-RFQ-API-Key": process.env.SODU_RFQ_API_KEY };
const baseUrl = process.env.SODU_BASE_URL;
await fetch(`${baseUrl}/api/rfq/orders/from-rfq`, { method: "POST", headers, body });
```